<u>WEEKLY REPORT :</u>

Project Title: Self Driving RC Car

Date: 9/8/2026

Instructor: Dr. Nguyen

TA: Ruben Ramirez

<u>Project Members:</u>

- Ethan Wells (Group Leader)
- Alexis Perez
- Ethan Bishop
- Abigail Duran

- This week the team moved into buying hardware and starting on the software side. We bought the Raspberry Pi Compute Module I/O Board, which came in this week, so we have a board to prototype the CM5 carrier design on. Abigail drew up the schematic Professor Nguyen asked for. We also ordered the rest of the core sensors: an ultrasonic sensor for measuring distance to obstacles and walls, an IMU for tracking heading and turns, and the camera that reads the traffic signs. Ethan Wells started researching how to train the traffic-sign model in TensorFlow, which is the framework the proposal calls for, and set up a GitHub repository to hold the training code. As a group we settled on standing weekly meeting times in the Robotics Lab in the Delta Building: Mondays 4:00 to 7:00 PM, Wednesdays 1:00 to 6:00 PM, and Thursdays 10:00 AM to 12:00 PM.

<u>Proposed Plan for Next Week:</u>

- <u>Ethan Wells:</u> Get a first training run going in TensorFlow on a public traffic-sign dataset (LISA), push the training script to the GitHub repo, and start figuring out how to convert the model to TensorFlow Lite so it can run on the CM5.

- Alexis Perez: Get the ultrasonic sensor in and test it on the bench, make sure it returns distance readings, and figure out its usable range and how it reads against a wall.
- Ethan Bishop: Get the IMU in and test it on the bench, make sure it reports heading and orientation, and check how noisy the readings are.
- Abigail Duran: Get the camera in and make sure it captures frames, and rework the schematic based on Professor Nguyen's feedback so it is ready to lay out the carrier board.

Weekly Contributions:

- Ethan Wells:
- 9/8/2026 – Bought the Compute Module I/O Board (it came in this week), started researching how to train the traffic-sign model in TensorFlow, and set up a GitHub repository for the training code (5 hours)
- Alexis Perez:
- 9/7/2026 – Researched ultrasonic sensor options and ordered one to measure distance to obstacles and walls (3 hours)
- Ethan Bishop:
- 9/7/2026 – Researched IMU options and ordered one to track heading and turns (3 hours)
- Abigail Duran:
- 9/7/2026 – Worked on the schematic Professor Nguyen asked for (2 hours)
- 9/8/2026 – Finished the schematic and ordered the camera (3 hours)
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Hour Tracker:

Name	Hours For the Week	Cumulative Hours
Ethan Wells	5	5
Alexis Perez	3	3

Ethan Bishop	3	3
Abigail Duran	5	5